

Life in coniferous forest

For animals and trees alike, life in boreal forest is dominated by the need to survive long and extremely cold winters. Animals that remain active in winter, such as wolves, need a constant supply of food simply to avoid freezing. Conifers are difficult to exploit for food, which means that animals that rely on them have developed some highly specialized physical and behavioral characteristics.

Feeding

Compared with many broad-leaved trees, conifers are well protected against attack. In addition to tough leaves, they often have oily resins that make both their leaves and their wood difficult to digest. Furthermore, if the sapwood is injured, this resin oozes out and traps insects and spiders as effectively as glue.

Despite these defenses, some animals manage to live entirely on coniferous trees. Among the most successful are sawfly larvae. These caterpillarlike grubs bore deep inside the trunks, leaving cylindrical tunnels in the wood. They do not eat the wood itself; instead, they feed on a fungus that grows on the walls of the tunnels they have built. Female sawflies carry small amounts of this fungus with them when they emerge as adults, and they infect new trees when they lay their eggs. This kind of mutualistic partnership is vital to sawflies, but it is not entirely unique: in other habitats, particularly in the tropics, ants and termites also “cultivate” fungi as food.

Although wood-boring grubs are safe from most predators, they are not entirely immune from attack. Coniferous forest is the habitat of some of the world’s largest woodpeckers, which hammer their way into tree trunks to reach the grubs inside. Sawflies also face a threat from the ichneumon wasp, which drills through the wood with its long ovipositor to

lay its eggs on the sawfly larvae. It is thought that the wasp locates the larvae by smell. When the eggs hatch, the ichneumon grubs eat their host larvae alive.

Some animals, such as the capercaillie and North American porcupine, eat large quantities of conifer needles, but moth caterpillars are the leading leaf eaters: as always in coniferous forest, the number of species involved is small, but the damage they inflict can be

EXTRACTING SEEDS

A crossbill's beak overlaps at its tips, making it an ideal tool for extracting cone seeds. For short periods, crossbill nestlings can survive temperatures as low as -31°F (-35°C).



vast. This is especially true of species such as the gypsy moth, which has been accidentally introduced to many parts of the world.

Finding winter food

Conifers do not have flowers, but they nevertheless produce seeds. For birds and small mammals, this seed crop is a valuable winter fuel. However, accessing conifer seeds is not easy: they develop inside woody cones, and the cones stay tightly closed until the seeds inside are mature.

Coniferous forest animals have developed a variety of ways of removing these seeds before the trees scatter them on the forest floor.

Squirrels gnaw through the soft, unripe cone while it is still attached to the branch, eating the seeds and dropping the remains of the cone on the ground below. Woodpeckers often take fallen cones and wedge them into tree holes or broken stumps, using these to hold the cone firm while they peck out all the seeds. Crossbills are even more proficient: their beaks are uniquely adapted for dealing with cones, enabling them to extract the seeds with surgical precision.

Compared with seeds, bark is a low-quality food, but in winter it is vital to some species' survival. Deer strip it away from the base of young saplings, while bank voles and porcupines often climb trees to attack the bark higher up. Bark stripping often stunts a tree's growth, and a severe attack can kill it.

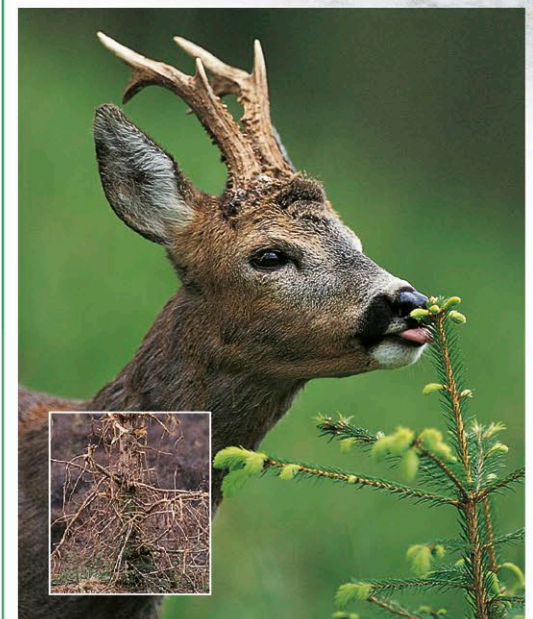
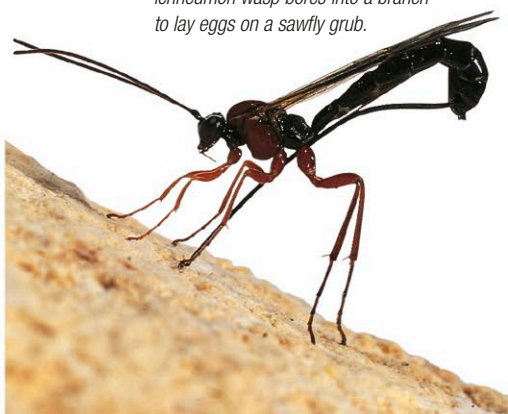


MOTH ATTACK

Although the gypsy moth itself does no damage, its caterpillars feed voraciously on conifers as well as on a range of deciduous trees.

DRILLING THROUGH WOOD

Bracing itself with its feet, a female ichneumon wasp bores into a branch to lay eggs on a sawfly grub.



BROWSING DEER

Deer have a major impact on coniferous forest by eating young tree shoots and bark. Where deer are numerous, such browsing can kill small saplings (inset). However, this creates open glades, which are useful to other animals because they let in light, allowing fruit-forming plants—a good food source—to grow.

Coping with cold

Coniferous forest has its share of seasonal visitors: principally insect-eating birds that arrive in spring and then leave for the south once they have raised their young. But for the rest of its animals, long winters are an inescapable fact of forest life. Some species hibernate, but many remain active even during the coldest months, relying on their insulation for survival.

One group of forest animals, mustelids, have coats that are exceptionally well insulating. This group includes pine martens, wolverines, minks, and sables, all of them agile hunters renowned for their thick and luxurious fur. As with most mammals, their coats contain two different kinds of hair: long, outer guard hairs